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Artificial Intelligence

Mid Activity #2

Documentation Explain Minimax in my own words:

The Minimax algorithm is a strategy used in two-player games like Tic-Tac-Toe that helps an AI choose the best possible move while minimizing the players'. Then the AI selects the move that maximizes its score while assuming the opponent will make moves that minimize it. In this way, the AI looks ahead at every possible outcome, predicts how the opponent will respond, and picks the move that provides the best guaranteed result. This makes Minimax a reliable method to ensure the AI never loses because it will always either win or force a draw with perfect play. Even if the player goes first, it will still never end up with the AI losing as MinMax will not allow it to lose a game, not even once.

Show Alpha-Beta Pruning Effect (Node Counts or Timing):

Alpha-beta pruning is a way to make the Minimax algorithm faster. It helps the computer skip checking some moves that it already knows will not lead to a better result. For example, without alpha-beta pruning, the computer might check out more moves than necessary and take more time to think. With alpha-beta pruning, it only checks the necessary moves and decides in less than a second. This makes the game run faster while still letting the computer play just as smart.

Reflection (One insight or challenge you faced):

One challenge I faced was understanding how the computer takes turns when thinking about the game. I was confused about how it knows when to choose the best move for itself and when to choose the worst move for the other player. I learned that the computer takes turns in its thinking, pretending to be both players one after the other. I also learned that alpha-beta pruning helps it think faster without making it any less smart.